CYBR 437 Secure Coding
Winter 2026
Lab - 10

Release date: March 10, 2026

Due date: March 16, 2026, at 11:59 pm

Introduction

This lab will focus on vulnerability assessment using open-source static-analysis tools. Install the following tools, discussed in lecture, to analyze the provided source code: `cppcheck`, `flawfinder`, `splint`, `clang static analyzer`. Review lecture slide deck 16 for instructions on installation and use. Use `flawfinder --context <sourcefilename>` for added details in the analysis output of `flawfinder`. In addition, you will also write your own mock-assessment tool using Regular Expressions to identify vulnerable/deprecated functions in C source files.

Task1: (20 points)

Analyze the source file `task1.c` with the 4 tools mentioned above and explain the outputs obtained from each of them. Also, provide a comparative analysis of the outputs. Include relevant screenshots of outputs in your write-up description.

Task 2: (30 points)

Write a small Python program that uses regex to detect the use of vulnerable/deprecated functions in C source files. The program should output a message for each occurrence of such a function, the line number where it occurs, and the line of code for the user. Use your analyzer to find the vulnerable functions in the source file `task2.c`.

Submission Instructions

1. Your write-up submission should include:
   - Detailed explanation of the analysis output for Task 1 with relevant screenshots
   - Detailed explanation of your Python code and relevant screenshots for Task 2

2. Python file (your code) for Task 2